

Improving the Classification of Arabic Tweet Sentiment Using Word Embedding Models

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Abstract—Enhancing the Classification of Sentiments in Arabic Tweets through Word Embedding Models Sentiment analysis in Arabic tweets is applied in this research using word embedding techniques. The primary objective is to determine how to classify Arabic tweets as positive or negative. The analysis was based on a dataset of 4,546 Arabic tweets obtained from Kaggle. The word embedding model (e.g. Word2Vec and GloVe) was used to facilitate the analysis because its semantic meaning and relations of words were better than what was provided through traditional methods like TF-IDF. To analyze the data, some of the classification techniques that were used include logistic regression, decision trees, and support vector machines. The results of the study showed that SVM has the best accuracy at 87%, followed by decision trees at 80%, and then logistic regression at 85%. This demonstrates that word embedding simulation models are effective for Arabic sentiment analysis on social networking sites. With 80% decision trees, word embedding greatly aided pyramid computing's divergence modulation, which allows for sentiment and mining analyses of Middle Eastern users' social activities. SVM performed well at the more sensitive and sophisticated tasks of opinion and market research analysis involving the Arabic language. Even so, a more precise classification can be performed with the application of deep learning algorithms or a hybrid model in future work.

Keywords—Sentiment Analysis of Arabic Text, Embedding of Words, Automatically Text Categorization, Support Vector Machine Classification, Natural Language Processing.

I. INTRODUCTION

This paper is devoted to three components, these are social media, sentiment analysis, and the Arabic Language. Social media has permeated the social structure of contemporary society, bringing with it the pressing need for mechanisms to gauge and detect sentiments. We would argue that there is an increase in Arabic content on many social networks. In [1] they noted, the sophisticated world of technology constantly drives the need for sharp instruments to tackle the Arabic Language, and the paper in [2] also notes that sentiment programs are increasing to “know” the Arabic language. The deep problem it suffers is that Arabic Language, like many languages, is multifaceted but in its case, it is based on the most profound cultural sociology of poem dialects and context-dependent vocab.

NLP sentiment models are unable to function efficiently due to the absence of reliable Arabic natural language

processing tools, which is one of the major issues that has plagued the region.

While traditional methods such as TF-IDF are effective, this research employs machine learning methods such as Word2Vec and GloVe. Additionally, sedentary methods like logistic regression, decision trees as well as support vector machines are implemented to classify Arabic tweets into positive or negative sentiments. The data set for this study was acquired from Kaggle which consists of 4,546 Arabic tweets[3].

This research intends to enhance the sentiment classification accuracy of the Arabic language through the application of word embedding models combined with machine learning techniques. Using datasets that are accessible on Kaggle, this study aims to find the most effective algorithm for performing sentiment analysis for tweets written in Arabic. Developing more effective sentiment analysis, market research tools, social public opinion monitoring tools, as well as social media analysis tools for use with the Arabic language-speaking population is one of the intended outcomes of this research.

II. LITERATURE REVIEW

Sentiment analysis also referred to as Opinion mining, is defined by [4], [5] as a sub-branch of natural language processing that involves the detection and classification of subjective information from a given text.

The rise in social media activities, especially micro-blogging activities on Twitter, has led to an increase in user-generated content. This development has made public sentiment analysis an essential tool for monitoring social media and emerging trends as noted by [6]. However, Arabic natural language processing is still very underdeveloped due to the highly diverse and complex nature of Arabic dialects.

In [7], they suggested a method called aspect-based sentiment analysis (ABSA). The method attempts to enable the system to identify which aspects of a document are referenced in each sentiment statement made. There are two principal steps in ABSA: aspect term extraction and aspect sentiment classification. While studying the Arabic language, in [2] they employed sense disambiguation using word embeddings as the primary representation of features. They built models, based on the assumption that syntactic and semantic relations in Arabic are richer than traditional NLP accounts for, and therefore should be approached with word

embedding. Some linguistic problems were solved using collecting parallel Arabic texts from different domains with the aid of Word2Vec models developed by [1].

Their study indicates that sentiment classification accuracy of 91% to 95% using an Arabic health sentiment dataset can be attained by incorporating a Convolutional Neural Network (CNN) with word embeddings. Findings emphasized the importance of some sets in a sentiment classifier's accuracy. To further this, in [8] they added word embeddings to modern architectures such as CNNs, LSTMs, and CNN-LSTM models. With FastText embeddings, His model achieved 94.69% and cited this hybrid model as important for sentiment analysis. In [9] they added to this argument claiming there is a high necessity for these models to be developed to accurately address the complexities of the Arabic language. Their work on ArWordVec provided evidence that such models can improve the performance and accuracy of more classical approaches. It was shown that ArWordVec, a new model for sentiment analysis of Arabic Twitter data, can outperform traditional systems, further demonstrating the advantages of modern methods for NLP problems. In conclusion, it appears that prior studies focused on sentiment analysis have employed deep learning techniques and sophisticated embeddings.

Nevertheless, one can argue that the traditional methods of Arabic text categorization are no longer useful due to the existence of a blend of new and old machine learning methods that is less costly to implement while highly effective. Newer studies have especially tried to define new methods in performing analysis of sentiment in Arabic due to its complex morphology, consonantal writing system, and the various dialects spoken around the world. Recent study [10] researched the impact of adjustable parameters within word embedding models and how they relate to the precision of Arabic sentiment analysis. These researchers demonstrated that improved classification accuracy could be achieved by adjusting parameters such as the window size, word vector dimensions, and negative sampling. In [11] they explain a novel technique of word embedding in Arabic based on The Bag of Roots approach combined with Global Vectors (GloVe). This approach, as the researchers emphasized, enhances the sentiment scoring process because words that have the same root contain similar emotions, and thus less time is needed for sentiment analysis on Arabic texts.

In [12], the work marks the highest-quality dataset available for Arabic twitter sentiment analysis[12]. Such a resource allows for further research and construction of classifiers that work at the more precise levels of the Arabic language and is now considered to be one of the main aids to the models that perform sentiment analysis for the structure of the language.

These developmental efforts show the endless innovations and automation that have been made in analyzing the sentiments in Arabic languages, especially with deep learning neural networks, context embeddings from words, and building comprehensive language-specific datasets and phrases. With the advancement of deep learning and word embeddings, the problem of analyzing sentiments in the Arabic language has been made easier. As mentioned earlier, the data's morphology and dialect-based structure shape the preprocessing of data. However, the real gap in research is much more complicated and stems from older versus newer machine learning systems and how they deal with small

datasets. Even though deep learning and the rest of modern models are known for being accurate, systems that are real-time powered with fewer computational resources tend to utilize classical methods that have very little reference in modern literature.

Compared to other methods, deep learning methods such as recurrent neural networks (RNN) and transforms have higher accuracy and lower interpretability [13]. Other forms of machine learning like logistic regression, decision trees, and support vector machines (SVMs) achieve reasonable accuracy with modest data sets and are easy to interpret [14].

In this context, this research focuses on a relatively unexplored gap in the literature that concerns the embedding of translation with the sentiment analysis of Arabic tweets. It focuses on the TF-IDF word representation accuracy and benchmarks the performance of multiple algorithms. Also, this study builds upon previous studies in a novel way by SVM's use in sentiment analysis classification tasks with traditional machine learning techniques so that this study did not only aim at enhancing previous studies but also showed that better classification results can be accomplished by the use of word embeddings.

Building upon the previous research has been challenging due to the recent advances in NLP techniques and the explosion of sentiment-annotated Arabic data. Most researchers attempted to deploy deep learning techniques, which are powerful but expensive and less efficient than other simpler models.

For Arabic sentiment analysis, datasets such as The Arab Tweets Corpus [15] and the Arabic Sentiment Dataset [16] available in Kaggle are particularly important. The research examines an Arabic tweet dataset sourced from Kaggle that is sufficient for sentiment analysis [3]. Similar to other studies, performance comparisons are made through evaluation accuracy, precision, and recall metrics which serve as criteria for performance assessment. Users' metadata and archival tweet information have for instance been used to enrich the classifications of tweet sentiments. In [17] they showed that adding a contextual layer to the user variable improved the performance of the classifier, thus enabling better comprehension of Arabic sentiment analysis.

III. PREPROCESSING TECHNIQUES

The effectiveness and accuracy of preprocessing significantly impact NLP tasks such as text classification and sentiment analysis. Key preprocessing steps include normalization, which standardizes text, and tokenization, which is particularly important for handling Arabic text. Additionally, stop words—frequent words that carry little meaningful information—should be removed, along with emojis, to ensure that sentiment analysis focuses solely on textual content without the influence of non-textual elements. These preprocessing techniques collectively enhance the overall performance of NLP models as shown in Table 1

Labeling	Number of tweets	Example	Example (cleaned data)
positive	2436	حقوق المرأة ❤️❤️❤️ https://t.co/Mzf90Ta5g1	حقوق مرأه حب حب حب
negative	2110	دوري الابطال اهم من حقوق المرأة	دور ابطال هم من حقوق مرأه

Table 1: Sample Dataset Before and After Preprocessing

IV. CLASSIFICATION METHODS

This evaluation focuses on Logistic Regression, Decision Trees, and Support Vector Machines. Regression analysis encompasses the logistics of the inter-relationship between a dependent variable and independent variables. Decision Trees allow for the efficient analysis of multiple possible outcomes to systematically aid in the decision-making process of dividing a dataset into multiple segments through a graph. Support Vector Machines are mainly used for pattern recognition processes for both classification and regression analyses.

Recent reports suggest these methodologies can be utilized on virtually anything. For instance, it has been shown that the predictive accuracy level of Support Vector Machines was always higher than that of the Logistic Regression models. For example, the subsequent analysis using the Support Vectors Machine and Logistic Regression for the more accurate methods of diagnosing obstructive sleep apnea enhanced research attempts far more than the Logistic Regression Method did. In a comparative study of machine learning text classification, almost all participants earned high accuracy rates while using both Support Vectors Machine and Logistic Regression, proving that these two algorithms are indeed effective and dominant in the classification field (Comparative Analysis of Machine Learning Algorithms).

The unmatched performing algorithms like Logistic Regression, Decision Trees, and Support Vectors Machines are quite effective as shown in the models above.

Let us try to explain everything step by step. First, metrics can be set differently so they can take care of a specific situation. Second, for prototypes that need to perform a specific tool set evaluation to get probability scores using the precision posters, an assessment has to be done. Third, contextual requirements, precision, and recall differences are supplied through the analysis. Fourth, task or activity scores are turned into precision wherein the average score of each class enrollment is taken into consideration for more context.

The averaging of the student's scores in a class determines the variance. After counting all subclasses, each distinct entry is counted to meet the support requirement. Support's unique values can also be fetched from each class. For multi-range queries, these unique values of support can be harvested later when students are selected. Milestone metrics are derived from the sum of support out of which baseline metrics are constructed. In addition to all previously mentioned grades metrics can also be created using multi-row selection criteria without the parameters set. For example, the maximum bound could set criteria within immediate boundaries such as mid-tier span or grade point.

These are two more ways the challenge is encountered, when accuracy does not work remaining precision and recall must undergo scrutiny. The latter concerns focus on how many and goal targets the values identified captured. The second same approach, where students might know something more refined than interpreted, where values are greater than expected.

The evaluation metrics are mathematically defined as follows:

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

$$Precision = TP / (TP + FP) \quad (2)$$

$$Recall/Sensitivity = TP / (TP + FN) \quad (3)$$

$$F1\ Score = 2 \times (Precision \times Recall) / (Precision + Recall) \quad (4)$$

Where: **TP**: True Positives, **TN**: True Negatives, **FP**: False Positives, **FN**: False Negatives

These metrics are fundamental in evaluating the performance of classification models, providing insights into their accuracy and effectiveness in real-world applications [18].

V. RESULT

Around 87% accuracy marks how well the Support Vector Machine (SVM) classifier performed compared to all other models as shown in Figure 1, which clearly illustrates its prominent effectiveness in sentiment classification, in particular, for Arabic tweets which have a profoundly complex linguistic structure. Whereas, Logistic Regression also performed well with an accuracy of nearly 85%. This model proves to be dependable for identifying binary classification approaches although it lacks the proficiency required for identifying more complex structures when pitted against SVM.

The lowest accuracy was given to the Decision Tree classifier at approximately 80%. As is the case with other complex overfitting datasets, overfitting ability paired with low generalization limits the usefulness of the model for sentiment classification across cross-linguistic situations.

Furthermore, normalizing, tokenizing, stemming, and the removal of stop words and reducing noise were apparent in them providing a better representation of text for a deeper understanding of sentiments. These techniques foster better outcomes for different NLP applications such as sentiment analysis, text classification, named entity identification, and machine translation.

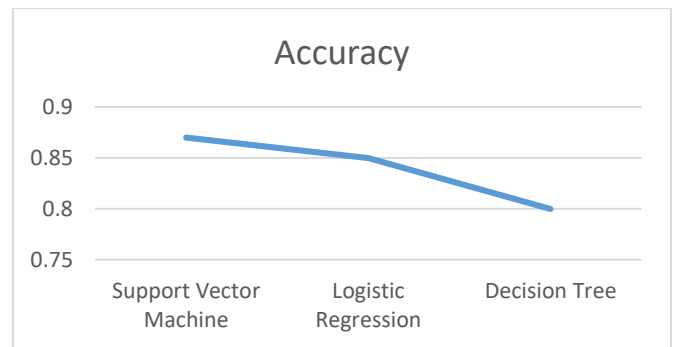


Figure 1: Accuracy Representation.

VI. CONCLUSION

The results highlight the effectiveness of SVM modeling especially when it comes to classifying sentiments in Arabic text, as it proves to be the most effective model in dealing with the linguistic ambiguity posed by Arabic. Logistic Regression achieves some desirable results but is still inadequate for fully capturing the nuances of sentiment-colored text. Decision

Trees, on the contrary, face difficulties in their ability to generalize because they overfit.

Furthermore, the classification accuracy is greatly improved by using good text preprocessing, which further exhibits its relevance in Arabic NLP. Research can be done on hybrid models or even deep learning algorithms so that accuracy in the classification tasks can be achieved. Also, further enhancing the customized preprocessing for the Arabic language can help in accomplishing more advanced configurations for sentiment analysis and other NLP applications.

VII. FUTURE WORK

One important area that can be pursued in the future is the incorporation of large tweet datasets and other social media content which might yield fresh and more productive insights into the model's performance.

Also, the ability of the model to better understand context and subsequently generalize can significantly improve by using word embedding software like FastText instead of TF-IDF. In addition, classification accuracy can be improved through better text representation since FastText excels in dealing with big data.

Subsequent research could take on different supervised approaches, like K-Nearest Neighbors (KNN), Naïve Bayes, or Linear Regression, as well as other diverse machine learning methods. In addition, the application of some unsupervised techniques such as K-Means, Hierarchical Clustering, or Density-Based Spatial Clustering of Applications with Noise (DBSCAN) can yield valuable insights into the sentiment and structure of the text.

Subsequent studies could improve the classification accuracy by attempting hybrid models or deep learning techniques. Incorporating more sophisticated neural architectures like RNNs or transformers could create more effective sentiment analysis systems especially when dealing with the Arabic language.

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